Optimization of Electric Vehicle Charging Infrastructure in Istanbul

Armin Ekberi
Department of Artificial Intelligence and Data Engineering
Istanbul Technical University
ekberi24@itu.edu.tr
150240703

İsmail Köylü
Department of Artificial Intelligence and Data Engineering
Istanbul Technical University
koylu23@itu.edu.tr
150230306

M. Ahmet Sadi Kerimi
Department of Artificial Intelligence and Data Engineering
Istanbul Technical University
kerimi23@itu.edu.tr
150230319

I. Project Description

This project aims to optimize the electric vehicle charging infrastructure in Istanbul. For this purpose, both the capacity adjustment of existing charging stations will be optimized, and new stations will be opened in candidate locations. Considering the increasing number of electric vehicles over the years, existing charging stations may become insufficient in different regions. To determine this demand increase, the total number of electric vehicles will be estimated based on the year selected in the slider. This estimated demand will then be distributed to demand zones.

The main objective of the project is to determine where additional charging capacity should be allocated and which candidate locations should have new stations opened, so that the system remains accessible, cost effective, and less dense. The final system will include interactive controls such as a year slider and a demand range slider, and will visualize station usage on a map using red circles with different opacity levels. Newly selected charging stations will be indicated with **X** marks.

II. Problem Definition

We formulate the EV charging infrastructure planning problem for Istanbul as a capacitated location assignment and capacity expansion model. Let D denote the set of demand regions, E the set of existing charging stations, and C the set of candidate locations for new stations. We define $S = E \cup C$ as the set of all charging service points.

For a selected year y, the total EV population is estimated by a growth function $N(y)$. To account for uncertainty, a demand scaling parameter θ is introduced through an interactive range slider. The charging demand of region d is defined as

$$q_d(y, \theta) = \theta N(y) w_d,$$

where w_d is the normalized demand weight of region d and $\sum_{d \in D} w_d = 1$.

The model determines how much additional capacity should be assigned to existing stations and which candidate locations should be opened as new charging stations. Demand from each region is assigned to available service points while unmet demand is penalized in the objective function.

Decision Variables

- $x_i \geq 0, \forall i \in E$: additional capacity assigned to existing station i
- $y_j \in \{0, 1\}, \forall j \in C$: equals 1 if candidate station j is opened
- $z_j \geq 0, \forall j \in C$: installed capacity at candidate station j
- $a_{ds} \geq 0, \forall d \in D, s \in S$: amount of demand from region d assigned to station s
- $u_d \geq 0, \forall d \in D$: unmet demand in region d
- $o_s \geq 0, \forall s \in S$: The overload variable measures the overload at the station s.

Objective Function

The objective is to minimize total accessibility cost, unmet demand, investment cost, and congestion:

$$\min \Bigg[\alpha \sum_{d \in D} \sum_{s \in S} c_{ds} a_{ds} + \beta \sum_{d \in D} u_d + \gamma \left(\sum_{i \in E} g_i x_i + \sum_{j \in C} f_j y_j + \sum_{j \in C} h_j z_j \right) + \lambda \sum_{s \in S} o_s \Bigg]$$

where c_{ds} denotes the Euclidean distance computed from the coordinates of demand region d and charging station s, g_i is the unit expansion cost of existing station i, f_j is the fixed opening cost of candidate station j, and h_j is the installation cost of capacity at candidate station j.

Main Constraints

The model has three main groups of constraints:

- **Demand satisfaction constraints:** regional demand must either be assigned to a charging station or remain as unmet demand.
- **Capacity and opening constraints:** assigned demand cannot exceed the available capacity of existing or newly opened stations, candidate stations can serve demand only if they are opened.
- **Budget and accessibility constraints:** the total expansion and opening cost must remain within the available budget, and demand can only be assigned to stations within a predefined maximum Euclidean distance.

In this formulation, the year slider and demand range slider act as hyperparameters that change the demand scenario, while the optimization model produces the corresponding capacity expansion and station opening decisions.

III. Dataset

The project will use real world datasets for Istanbul collected primarily from the Istanbul Metropolitan Municipality Open Data Portal and the National Smart City Open Data Platform. The main datasets will include (i) existing EV charging station location data, (ii) charging station socket data to represent current station capacity, (iii) ISPARK parking lot data to define candidate locations for new stations, and (iv) district level demographic data to construct regional demand weights. In addition, a year based EV growth function will be derived from official EV related statistics published by public institutions such as EPDK and TÜİK. Accessibility cost will be modeled using Euclidean distance between demand region centroids and charging station locations. These datasets will be integrated to estimate regional charging demand and optimize both capacity expansion at existing stations and opening decisions at candidate locations.

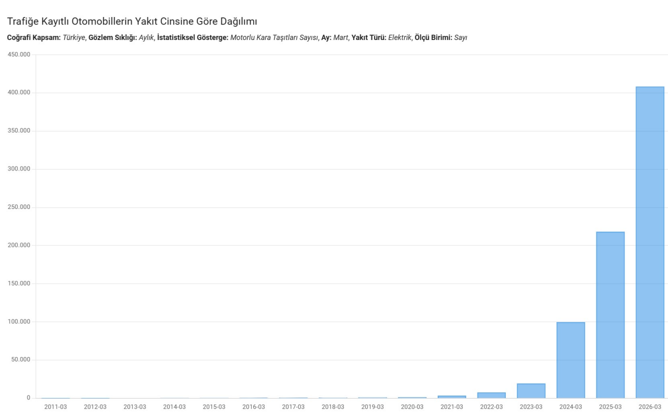

Fig. 1. Electric vehicle growth trend in Turkey by year. This trend will be used to estimate the year based EV demand function $N(y)$ in the proposed model. (TÜİK)

[5].

IV. Methodology

The proposed problem will be modeled as a capacitated location assignment and capacity expansion problem and solved using a mixed integer linear programming (MILP). The methodology has four main stages. First, a year based EV growth function $N(y)$ will be estimated from official EV statistics in order to approximate the total EV population for the selected year. Second, this total demand will be distributed across demand regions using normalized district level demand weights, which will generate the regional charging demand values used in the model. Third, the accessibility cost matrix will be constructed by computing Euclidean distances between the centers of demand regions and the coordinates of charging stations. These distances will be used in the objective function to represent accessibility in a computationally simple and clear way. Finally, the optimization model will determine the additional capacity assigned to existing stations and the candidate locations to be opened under budget, capacity, and accessibility constraints.

To evaluate different planning conditions, the notebook will include interactive controls such as a year slider and a demand range slider. The selected year will determine the base EV demand through $N(y)$, while the demand range parameter will scale this value to represent alternative scenarios. After solving the optimization model, the results will be visualized on a map. Station utilization levels will be shown by red circles with different opacity values, and newly opened stations will be indicated with X markers. This methodology is suitable because it combines real geographical data, optimization based decision making, scenario analysis through hyperparameters, and visual analysis of the resulting infrastructure plan.

References

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[3] https://ulasav.csb.gov.tr/dataset/34-ispark-otopark-listesi-web-servisi
[4] https://ulasav.csb.gov.tr/dataset/34-nufus-bilgileri
[5] Turkish Statistical Institute (TÜİK), "Distribution of Registered Automobiles by Fuel Type,"

V. GitHub Repository

Project Repository (Click)